赵齐 个人简历

基本信息

赵齐，男，31 岁，中山大学生物信息学博士，肿瘤防治中心博士后。长期从事生物信息学研究工作，在生物信息学模型构建和软件平台开发、肿瘤基因组大数据分析等方面积累了丰富的研究基础。先后在“天河二号”超级计算机上搭建了大数据建模与挖掘平台、超大规模肿瘤基因组数据分析平台，作为骨干全程参与肿瘤医院生物信息平台建设。利用上述平台完成近千例肿瘤组学高通量数据的分析工作，具有丰富的软件平台开发以及组学数据分析经验。学术方面，近些年在包括 ***Nature Materials, Nature Biotechnology, Cell research***， ***Nucleic Acid Research*** 等国际著名杂志上发表 SCI 论文 23 篇，其中主要作者文章 4 篇（见发表论文部分），总引用 451 次。其中，第一作者论文 GPS-SUMO（***Nucleic Acid Research***，IF=11.56）入选 ESI 高被引论文 Top 3%，SCI 引用 171 次。此外，作为共同第一作者首次揭示小细胞食道癌的基因图谱特征（***Cell Research***，IF=15.39），为相关治疗和药物开发重要依据。博士期间获国家研究生奖学金、有害生物控制国家重点实验“杰出研究生”一等奖，博士后期间获国家自然科学基金青年科学基金资助和中国博士后科学基金面上资助一等资助。独立/合作开发并发布开源生物信息学软件 10 余个，获颁软件著作权 3 项。

研究方向

1. 生物信息学算法模型构建及软件开发。
2. 消化道肿瘤组学数据分析及分子标志物筛选。

教育经历

1. 2005 年 9 月 – 2009 年 7 月 生物工程，工学 学士，陕西科技大学
2. 2011 年 9 月 – 2016 年 7 月 生物信息学，理学 博士，中山大学，导师：任间 教授

工作经历

2016 年 9 月 – 今 博士后， 中山大学肿瘤防治中心，合作导师：徐瑞华 教授

基金项目

主持

1 国家自然科学基金青年科学基金项目，81802438，*FAM83H-AS1* 在结直肠肿瘤中的功能和机制研究，2019/01-2021/01，21 万，在研，主持。

2 博士后科学基金面上资助，2017M0573，基于超级计算的肿瘤进化分析平台构建 2017/05-2018/09，8 万（一等资助），在研，主持。

参加

1. 国家自然科学基金面上项目，6187239，整合 DNA 三维结构信息的癌症基因组突变位点研究，2019/01-2022/12，65 万，在研，第一参与人。

2. 国家自然科学基金面上项目，81872011，*PDE3A* 促进食道小细胞癌发生发展的机制和靶向治疗研究，57 万，在研，第一参与人。
3. 广州市健康医疗协同创新重大专项，201510191740058，以瘤种为主题的生物医学大数据平台建设及精准医疗应用研究，2016/01-2018/12，77/600 万，在研，参与。
4. 国家自然科学基金面上项目，31471252，基于双分子荧光互补技术的小类泛素修饰因子调控机制的系统研究，2015/01-2018/12，85 万，在研，参与。

主要作者论文（#第一作者，*通讯作者）

1. **Qi Zhao**#, Yu Sun#, Dawei Wang#, Hongwan Zhang, Kai Yu, Jian Zheng*, Zhixiang Zuo*: LncPipe: A Nextflow-based pipeline for identification and analysis of long non-coding RNAs from RNA-Seq data. ***Journal of Genetics and Genomics*** 07/2018; 45(7).

2. Feng Wang#,*, Dong-Bing Liu#, **Qi Zhao**#, Gang Chen#, Xin-Ming Liu, Ying-Nan Wang, Hong Su, Yan-Ru Qin, Yi-Fu He, Qing-Feng Zou, Yan-Hui Liu, You-En Lin, Ze-Xian Liu, Jin-Xin Bei, Rui-Hua Xu*: The genomic landscape of small cell carcinoma of the esophagus. ***Cell Research*** 05/2018; 28(7).（共同第一作者）

3. Yubin Xie#, Xiaotong Luo#, Zhihao He#, Yueyuan Zheng, Zhixiang Zuo, **Qi Zhao***, Yanyan Miao*, Jian Ren*: VirusMap: A visualization database for the influenza A virus. ***Journal of Genetics and Genomics*** 04/2017; 44(5).（共同通讯）

4. **Qi Zhao**#, Yubin Xie#, Yueyuan Zheng#, Shuai Jiang, Wenzhong Liu, Weiping Mu, Zexian Liu, Yong Zhao, Yu Xue, Jian Ren*: GPS-SUMO: A tool for the prediction of sumoylation sites and SUMO-interaction motifs. ***Nucleic Acids Research*** 05/2014; 42(W1).

其他合作论文

1. Feng Ye, Hongwei Gao, Lin Xiao, Zhixiang Zuo, Yueping Liu, **Qi Zhao**, Huijiao Chen, Weiyi Feng, Bo Fu, Linyong Sun, Xiaolin Jiang, Du He, He Jiang, Mei Yang, Li Li, Fei Chen, Xiang Liu, Shuang Li, Zhihui Li, Yong Jiang, Liang Cheng, Hong Bu: Whole exome and target sequencing identifies MAP2K5 as novel susceptibility gene for familial non-medullary thyroid carcinoma: MAP2K5 is a novel susceptibility gene for FNMTC. ***International Journal of Cancer*** 08/2018;

2. Ying-nan Wang, Zhao-lei Zeng, Jiahuan Lu, Yun Wang, Ze-xian Liu, Ming-ming He, **Qi Zhao**, Zi-xian Wang, Ting Li, Yun-xin Lu, Qi-nian Wu, Kai Yu, Feng Wang, Heng-Ying Pu, Bo Li, Wei-hua Jia, Ming shi, Dan Xie, Tie-bang Kang, Peng Huang, Huai-qiang Ju, Rui-hua Xu: CPT1A-mediated fatty acid oxidation promotes colorectal cancer cell metastasis by inhibiting anoikis. ***Oncogene*** 07/2018;,

3. Yun-Xin Lu, Huai-Qiang Ju, Ze-Xian Liu, Dong-Liang Chen, Yun Wang, **Qi Zhao**, Qi-Nian Wu, Zhao-lei Zeng, Hai-Bo Qiu, Pei-Shan Hu, Zhi-Qiang Wang, Dong-Sheng Zhang, Feng Wang, Rui-Hua Xu: ME1 Regulates NADPH Homeostasis to Promote Gastric Cancer Growth and Metastasis. ***Cancer Research*** 04/2018; 78(8):1972-1985.

4. Qingbin Zhang, Li Chen, Shiman Cui, Yan Li, **Qi Zhao**, Wei Cao, Shixiang Lai, Sanjun

Yin, Zhixiang Zuo, Jian Ren: Expression and regulation of long noncoding RNAs during the osteogenic differentiation of periodontal ligament stem cells in the inflammatory microenvironment. ***Scientific Reports*** 12/2017; 7(1).

5. Rui-hua Xu, Wei Wei, Michal Krawczyk, Wenqiu Wang, Huiyan Luo, Ken Flagg, Shaohua Yi, William Shi, Qingli Quan, Kang Li, Lianghong Zheng, Heng Zhang, Bennett A. Caughey, **Qi Zhao**, Jiayi Hou, Runze Zhang, Yanxin Xu, Huimin Cai, Gen Li, Rui Hou, Zheng Zhong, Danni Lin, Xin Fu, Jie Zhu, Yaou Duan, Meixing Yu, Binwu Ying, Wengeng Zhang, Juan Wang, Edward Zhang, Charlotte Zhang, Oulan Li, Rongping Guo, Hannah Carter, Jian-kang Zhu, Xiaoke Hao, Kang Zhang: Circulating tumour DNA methylation markers for diagnosis and prognosis of hepatocellular carcinoma. ***Nature Materials*** 10/2017; 16(11).

6. F-H. Wang, C. Ren, Y. Zhang, S. Yao, H. Feng, H. Wu, H-F. Song, R. Zhang, X-L. Wei, X-J. Xia, Q. Zhao, J-P. Yun, B-Y. Zou, M. Qiu, Z-Q. Wang, Y-H. Li, R. Xu: 1186PPhase Ia study of a humanized anti-PD-1 monoclonal antibody (JS001) in Chinese patients with refractory solid tumors. ***Annals of Oncology*** 09/2017; 28(suppl_5).

7. Zai-Shang Li, Chuang-Zhong Deng, Yun-Lin Ye, Kai Yao, Sheng-Jie Guo, Jie-Ping Chen, Yong-Hong Li, Zi-Ke Qin, Zhuo-Wei Liu, Bin Wang, **Qi Zhao**, Peng Chen, Qi-Wu Mi, Xiao-Feng Chen, Hui Han, and Fang-Jian Zhou: More precise prediction in Chinese patients with penile squamous cell carcinoma: Protein kinase CK2α catalytic subunit (CK2α) as a poor prognosticator. ***Oncotarget*** 05/2017; 8(31).

8. Jinwen Feng, Chen Ding, Naiqi Qiu, Xiaotian Ni, Dongdong Zhan, Wanlin Liu, Xia Xia, Peng Li, Bingxin Lu, **Qi Zhao**, Peng Nie, Lei Song, Quan Zhou, Mi Lai, Gaigai Guo, Weimin Zhu, Jian Ren, Tieliu Shi, Jun Qin: Firmiana: Towards a one-stop proteomic cloud platform for data processing and analysis. ***Nature Biotechnology*** 05/2017; 35(5-5):409-412.

9. Guangyan Kan, Heng He, **Qi Zhao**, Xiubo Li, Min Li, Huasheng Yang, Jong Kyong Kim: Functional dissection of the role of UHRF1 in the regulation of retinoblastoma methylome. ***Oncotarget*** 04/2017; 8(24).

10. Long Bai, Feng Wang, Zhe-zhen Li, Chao Ren, Dong-sheng Zhang, **Qi Zhao**, Yun-xin Lu, De-shen Wang, Huai-qiang Ju, Miao-zhen Qiu, Zhi-qiang Wang, Feng-hua Wang, Rui-hua Xu: Chemotherapy plus bevacizumab versus chemotherapy plus cetuximab as first-line treatment for patients with metastatic colorectal cancer: Results of a registry-based cohort analysis. ***Medicine*** 12/2016; 95(51):e4531.

11. Zai-Shang Li, Chuang-Zhong Deng, Kai Yao, Yong Tang, Nan Liu, Peng Chen, Bin Wang, Xiang Li, Xiao-Feng Chen, Hong Liao, Qi-Wu Mi, Yong-Hong Lei, **Qi Zhao**, Pei-Zhen Zhao, Xue-Ying Li, Jie-Ping Chen, Qiang-Hua Zhou, Zi-Ke Qin, Zhuo-Wei Liu, Yong-Hong Li, Yun-Lin Ye, Hua Tu, Zi-Jun Zou, Xing Bi, Feng Yang, Ying-Ming Xiao, Jing Li, Xiang-Tian Lin, Wei-Cong Liang, Hui Han, Fang-Jian Zhou: Bilateral pelvic lymph node dissection for Chinese patients with penile cancer: a multicenter collaboration study. ***Journal of Cancer Research and Clinical Oncology*** 10/2016; 143(2).

12. Hui-Yun Liu, **Qi Zhao**, Tian-Peng Zhang, Yue Wu, Yun-Xia Xiong, Shi-Ke Wang, Yuan-Long Ge, Jin-Hui He, Peng Lv, Tian-Miao Ou, Jia-Heng Tan, Ding Li, Lian-Quan Gu, Jian Ren, Yong Zhao, Zhi-Shu Huang: Conformation Selective Antibody Enables Genome Profiling and Leads to Discovery of Parallel G-Quadruplex in Human Telomeres. ***Cell Chemical Biology*** 10/2016; 23(10).

13. Yubin Xie, Yueyuan Zheng, Hongyu Li, Xiaotong Luo, Zhihao He, Shuo Cao, Yi Shi, **Qi Zhao**, Yu Xue, Zhixiang Zuo, Jian Ren: GPS-Lipid: A robust tool for the prediction of multiple lipid modification sites. ***Scientific Reports*** 06/2016; 6:28249.

14. Haiying Liu, Qianqian Liu, Yuanlong Ge, **Qi Zhao**, Xiaohui Zheng, Yong Zhao: HTERT promotes cell adhesion and migration independent of telomerase activity. ***Scientific Reports*** 03/2016; 6:22886.

15. Zai-Shang Li, Peng Chen, Kai Yao, Bin Wang, Jing Li, Qi-Wu Mi, Xiao-Feng Chen, **Qi Zhao**, Yong-Hong Li, Jie-Ping Chen, Chuang-Zhong Deng, Yun-Lin Ye, Ming-Zhu Zhong, Zhuo-Wei Liu, Zi-Ke Qin, Xiang-Tian Lin, Wei-Cong Liang, Hui Han, Fang-Jian Zhou: Development of a new outcome prediction model for Chinese patients with penile squamous cell carcinoma based on preoperative serum C-reactive protein, body mass index, and standard pathological risk factors: The TNCB score group system. ***Oncotarget*** 03/2016; 7(15).

16. Zai-Shang Li, Kai Yao, Yong-Hong Li, Jie-Ping Chen, Chuang-Zhong Deng, **Qi Zhao**, Peng Chen, Bin Wang, Qi-Wu Mi, Zhuo-Wei Liu, Zi-Ke Qin, Hui Han, Fang-Jian Zhou: Clinical significance of preoperative C-reactive protein and squamous cell carcinoma antigen levels in penile squamous cell carcinoma patients. ***BJU International*** 11/2015; 118(2).

17. Wenzhong Liu, Yubin Xie, Jiyong Ma, Xiaotong Luo, Peng Nie, Zhixiang Zuo, Urs Lahrmann, **Qi Zhao**, Yueyuan Zheng, Yong Zhao, Yu Xue, Jian Ren: IBS: An illustrator for the presentation and visualization of biological sequences. ***Bioinformatics*** 06/2015; 31(20):3359-3361., DOI:10.1093/bioinformatics/btv362

18. Chao Shi, Jian Yin, Zhe Liu, Jian-Xin Wu, **Qi Zhao**, Jian Ren, Nan Yao: A systematic simulation of the effect of salicylic acid on sphingolipid metabolism. ***Frontiers in Plant*** Science 03/2015; 6:186.

19. Zhe-Zhen Li, Feng Wang, Zi-Chen Zhang, Fang Wang, **Qi Zhao**, Dong-Sheng Zhang, Feng-Hua Wang, Zhi-Qiang Wang, Hui-Yan Luo, Ming-Ming He, De-Shen Wang, Ying Jin, Chao Ren, Miao-Zhen Qiu, Jian Ren, Zhi-Zhong Pan, Yu-Hong Li, Jiao-Yong Shao, Rui-Hua Xu: Mutation profiling in Chinese patients with metastatic colorectal cancer and its correlation with clinicopathological features and anti-EGFR treatment response. ***Oncotarget*** 11/2014; 7(19).

软件著作权（作为主要完成人）

1. 图形化局部序列比对软件，2014SR004801，2014，中国，任间、张冉、谢宇斌、赵齐
2. 生物序列特征图标绘制软件，2014SR036498，中国，任间、刘文忠、赵齐、谢宇斌
3. 生物信号通路示意图绘制软件，2014SR038082，中国，任间、刘文忠、谢宇斌、赵齐

荣誉获奖

1. 国家奖学金（博士研究生），教育部（2015）
2. 杰出研究生“一等奖”，有害生物控制国家重点实验室(2014)

软件及数据库

1. GPS-SUMO: http://sumosp.biocuckoo.org
2. IBS: http://ibs.biocuckoo.org
3. IDEA: https://github.com/likelet/IDEA; http://idea.renlab.org
4. LncPipe: https://github.com/likelet/LncPipe
5. CrossICC: https://github.com/bioinformatist/CrossICC
6. VolcanoP: https://zhaoqi.shinyapps.io/VolcanoP/
7. DAtools: https://github.com/likelet/DAtools
8. PlotAppForTBtools: https://github.com/likelet/PlotAppForTBtools
9. lassBag: https://github.com/likelet/lassoBag
10. VirusMap: http://virusmap.renlab.org/en/index.php

其他个人信息

1. 个人主页：https://likelet.github.io
2. ResearcherID: http://www.researcherid.com/rid/M-4830-2016
3. Github website: https://github.com/likelet
4. ResearchGate: https://www.researchgate.net/profile/Qi_Zhao11
5. ORCID: https://orcid.org/0000-0002-8683-6145
6. Google Scholar: https://scholar.google.com/citations?user=j7LCthMAAAAJ&hl=en